

Scenario, Role, and Persona: A Scoping Review of Design Strategies for Socially Intelligent AI Agents

Ruowen Niu*

Hong Kong Generative AI Research
and Development Center
Hong Kong SAR, China
verityniu@ust.hk

Jiaxiong Hu*

Department of Computer Science and
Engineering
The Hong Kong University of Science
and Technology
Hong Kong SAR, China
hujx@ust.hk

Siyu Peng

The Hong Kong University of Science
and Technology
Hong Kong SAR, China
spengai@connect.ust.hk

Caleb Chen Cao

Big Data Institute
Hong Kong University of Science and
Technology
Hong Kong SAR, China
cao@ust.hk

Chengzhong Liu

Hong Kong Generative AI Research
and Development Center
Hong Kong SAR, China
chengzhong.liu@connect.ust.hk

Sirui Han†

The Hong Kong University of Science
and Technology
Hong Kong SAR, China
siruihan@ust.hk

Yike Guo†

The Hong Kong University of Science
and Technology
Hong Kong SAR, China
yikeguo@ust.hk

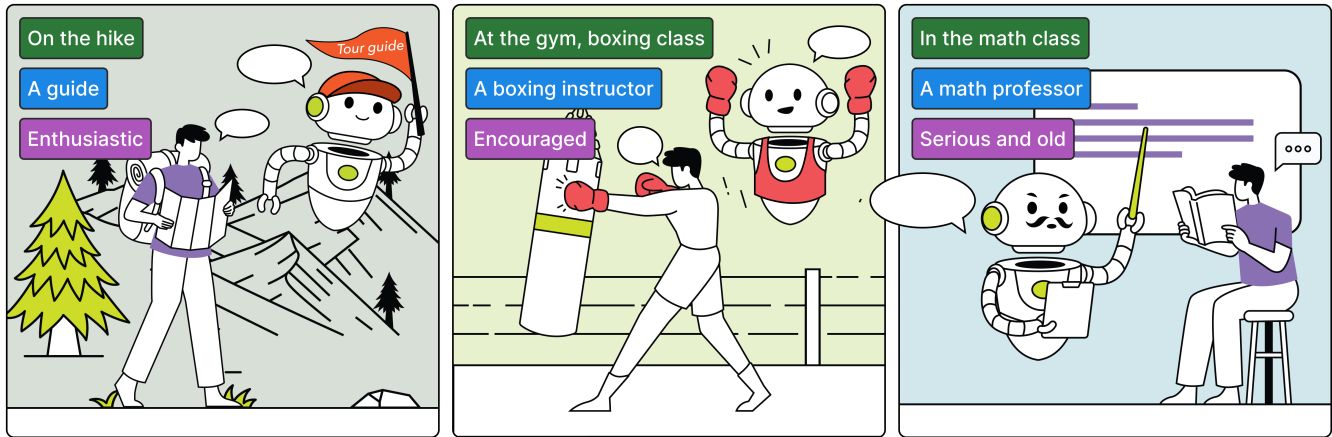


Figure 1: Examples of AI agent design showcasing the strategy implications of scenario (green label), role (blue label), and persona (purple label).

*Both authors contributed equally to this research.

† Corresponding author

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Abstract

As artificial intelligence (AI) agents have often been perceived as social actors, there is a growing expectation for them to demonstrate social intelligence. Social intelligence encompasses the ability of AI agents to align their actions with human intentions and behave in socially and culturally appropriate ways. This study conducts a scoping review to analyze the design strategies employed in current AI agent designs that support the exhibition of social intelligence. Our findings reveal three design strategy themes that interdependently and collectively shape the capabilities and behaviors of AI agents:

scenario, role, and persona. These findings provide a structured perspective and actionable insights for designing AI agents that effectively integrate social intelligence into human-AI interactions.

CCS Concepts

• **Human-centered computing** → **HCI theory, concepts and models.**

Keywords

AI Agents; Design Strategy; Scoping Review; Social Intelligence

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1 Introduction

Humans often subconsciously perceive artificial intelligent (AI) agents as social actors during interactions [40], reflecting an expectation that AI agents exhibit “social smarts” [9, 37]. This expectation highlights the need for AI agents to understand human intentions and align with social norms [49], thereby positioning social intelligence as a key design requirement in human-AI interactions [15, 17]. Specifically, social intelligence refers to the ability to get along well with others and get them to cooperate in a certain task, encompassing understanding of others’ emotions and needs, the ability to adapt to different social contexts, and appropriate responsiveness [21]. Social intelligence allows AI to build a more positive relationship with humans by behaving in a way that aligns with user intentions and expectations shaped by social norms.

Current AI agents can display sufficient social cues to evoke the perception of social intelligence in users [3], while the design goals for achieving social intelligence in AI agents are highly application domain-specific [14, 15], as social norms and interaction requirements vary significantly across different domains. For example, in customer service, users expect professional interactions characterized by accurate and practical information, requiring AI agents to understand customer needs and demonstrate expertise along with skillful conversational techniques [13, 42]. To meet these expectations, designers can adjust the AI agent’s anthropomorphic role, interactivity, and appearance [13]. Similarly, in delivering reminiscence therapy for individuals with dementia, using appropriate language to evoke memories is the crucial requirement [8, 62]. Allowing users to select the agent’s preferred voice and gender helps establish an initial positive relationship with them. Additionally, incorporating personalized questions based on the user’s photos can effectively stimulate memories, enhancing the therapy’s engagement and impact [62]. These examples illustrate how various design strategies are applied to tailor AI agents with social intelligence to specific application domains, ensuring their interactions align with user expectations and domain-specific social norms.

To comprehensively explore various design strategies for the AI agents exhibiting social intelligence, this study conducts a scoping review, collecting and analyzing recent HCI studies on AI agents in

different application domains. Based on the analysis, we propose three themes of design strategies: *scenario*, *role*, and *persona*. Based on the three themes, we systematically discuss the implications of different strategies in specific application contexts. These themes provide insights into how design strategies can align the AI agent with social norms and human expectations to foster the socially intelligent AI agent.

2 Background

2.1 Related Sociological Theories

The Computers-Are-Social-Actors (CASA) paradigm posits that people apply the same social rules to technological artifacts as they do to human interactions [19, 40, 41] — suggests that individuals instinctively perceive AI agents as social actors, expecting them to adhere to social norms, demonstrate appropriate behaviors, and engage in meaningful interactions as similar as human. CASA provides a foundational understanding of why human social expectations shape interactions with AI. To unpack these expectations, sociological theories provide valuable frameworks for understanding how AI agents can effectively emulate social intelligence.

The dramaturgical theory provides a framework to understand the interaction and expectation among human [20]. Introduced by Goffman [20], this theory conceptualizes social interactions as theatrical performances, where individuals manage impressions by enacting roles appropriate to the social scenario. The performance that happens in certain scenarios encompasses behavior, how the role is enacted, and appearance, including clothing and physical presentation [20]. In the context of theatrical performances, roles are not just specific characters played by actors. Instead, they are sets of behaviors associated with particular social positions, constructed through patterned actions [5, 6]. The dramaturgical theory identifies two types of rules that govern behavior: normative rules, which dictate what is socially appropriate, and procedural rules, which ensure actions are predictable and consistent within a role [50]. Flexibly applying these rules in contextually appropriate ways demonstrates social competence and fosters trust, as it signals alignment with societal expectations and good character [50]. Failing to adhere to these rules can violate social expectations, disrupting the relationship among actors and undermining trust [7]. Applying this understanding to human-AI interaction, it is crucial for AI agents to be designed to fulfill human social expectations to ensure natural and effective interaction, fostering trust and positive relationships.

2.2 Social Intelligence in AI Agents

An AI agent’s social intelligence refers to how it can effectively and adaptively accomplish relevant objectives in specific social settings [18]. It makes the agent more than a simple tool to provide functions credibly, but to be socially and culturally acceptable [49]. Agents’ social intelligence plays a critical role in fostering trust and positive relationships in human-agent collaboration [46, 60] by meeting user expectations. Just as human behaviors in social contexts are guided by the needs and expectations of others [1], AI agents should align their actions with human intentions and expectations. For example, in learning settings, when users seek expert knowledge, an agent designed with formal communication,

the mature persona of a 40-year-old professor, and visible displays of expertise can foster trust and improve learning outcomes; when users desire peer encouragement and motivation, agents with a 20-year-old peer persona and casual interaction styles prove more effective [4]. Understanding and adapting behaviors to align with human intentions, as well as adhering to expected social norms, shows central expectations for social intelligence in AI agents. These expectations shape the design of AI agents, which should account for their intended use, the social environment's complexity, and the interaction's sophistication [17, 46]. Understanding these expectations is critical for formulating effective design strategies that enable AI agents to demonstrate social intelligence.

In summary, sociological theories not only provide a unique lens for understanding human interactions but also offer valuable insights for designing AI agents. However, how they are involved in the AI agent design process is unclear. By conducting a scoping review of existing AI agent design strategies, we aim to examine how these strategies align with sociological principles to meet user expectations. This analysis seeks to uncover the rationale behind these designs from a social intelligence perspective and develop practical insights to guide future designers in creating socially intelligent AI agents.

3 Method

AI agents — intelligent systems that autonomously interact with users and their environments, emulating human roles to deliver information across various applications — are investigated because their interactive nature necessitates task-specific functionality and the ability to demonstrate social interactions. These agents encompass embodied agents, robots, chatbots, and voice assistants. We adopted a scoping review approach as it aligns with our research objective of identifying the pattern of design strategies in AI agent design from the HCI perspective [39]. This scoping review focuses exclusively on the CHI '24 proceedings, a comprehensive platform that consistently integrates AI agent designs across diverse application domains. It offers an up-to-date perspective that bridges technologies and user experiences, making it an effective database for exploring specific topics within the HCI community, as highlighted in previous works [51]. We followed the PRISMA-ScR [57] approach to ensure rigor.

3.1 Collecting Papers

Building on search terms commonly used in prior literature reviews on AI agents [33, 65], we generated an initial set of keywords and tested them within the CHI '24 proceedings. Through an iterative process of refinement and incorporating additional keywords identified during testing, we finalized the search terms as ["AI" OR "Intelligent" OR "LLMs" OR "Large Language Models"] AND ["Agent" OR "Chatbot" OR "System" OR "Persona" OR "Robot"]. We applied these terms to search the paper abstracts across the selected data source. (See figure 2)

For the returned search result ($N = 77$), we first carried out a loose screening by reading the titles and abstracts of the papers to include those about AI agents, resulting in 24 papers being excluded. Next, we applied a stricter filtering process to the 53 papers, which involved reading the full article to determine if it met the following

criteria: (1) Articles that didn't position themselves as agents, such as report generators or writing tools, are excluded. (2) Articles that did not provide user-oriented interaction design contributions, such as solely evaluating performance and acceptance, are excluded. (3) Only complete, peer-reviewed, full papers are eligible for inclusion. Short papers, extended abstracts, and posters are excluded. Thirty papers are eligible for the analysis phase.

3.2 Analyzing Papers

This study employed an Abductive Thematic Analysis approach inspired by Thompson [55] to analyze the design strategies of AI agents. Abductive Analysis is particularly effective in forming explanatory hypotheses, which allows researchers to look beyond the data and the immediately observed patterns to consider broader implications and potential explanations [56]. By integrating theoretical insights from sociological theories, we ensured that the emerging themes were not only grounded in existing theories but also open to new data-driven discoveries [55]. This approach helped prevent the generation of arbitrary themes while allowing unexpected patterns to emerge inductively.

Following this methodological foundation, we adopted a structured coding process to identify patterns in AI agents design strategies. Two authors began the process by thoroughly familiarizing the data, followed by open coding of design strategies mentioned in articles, such as the design, method, and procedure section. To ensure consistency and reliability in the coding process, two authors independently analyzed 15 articles, listed the codes, and discussed them together to discover underlying patterns that could explain the data. Guided by dramaturgical theories, the analysis was structured using an initial codebook comprising three themes: *scenario*, *role*, and *persona*. The two researchers reviewed the remaining articles based on the agreement of the codebook. The final themes represent a synthesis of theory-driven categories and data-driven insights, providing a robust foundation for understanding the design strategies of socially intelligent AI agents across various domains.

4 Findings

Theories from sociology emphasize the environment that the individual is situated in, the role that the individual plays, and the personality, as well as the appearance, that the individual has for social interactions [5, 20]. These perspectives are also reflected in the studies we analyzed. We present the three themes (see table 2 in Appendix for detailed codes and themes), *scenario*, *role*, and *persona*, and propose design implications for the future design of AI agents with social intelligence in this section.

4.1 Scenario, Role, and Persona

These three themes are interdependent, shaping the agent's capabilities and behaviors to ensure alignment with user intentions and adherence to social norms, collectively constructing agents that not only fulfill functional objectives but also exhibit social intelligence effectively. Considering a Potions class at Hogwarts (see figure 3), the Potions classroom serves as the scenario, while the teacher and students represent the roles within this scenario. The Potions classroom establishes that the teacher's role is to impart knowledge about potion-making, while students are expected to

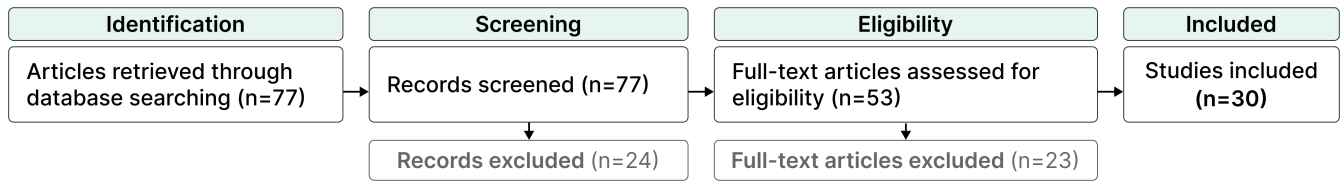


Figure 2: Flow diagram of the paper selection process.



Figure 3: Screenshot from Harry Potter, released by Warner Bros. Scenario: Potions class; role: teacher and student; persona: characters who have distinct stories and personalities, such as Hermione Granger, Harry Potter, and Professor Severus Snape

complete potion-related tasks. Any behavior unrelated to potion-making, such as off-topic activities, would be deemed inappropriate and result in deducting House Points. Within this scenario, there are multiple student roles, but each demonstrates unique personas, which define their behaviors and create lasting impressions. For instance, Hermione’s persona is characterized by her eagerness to answer questions and her academic excellence. In contrast, Harry’s persona reflects his dislike for Potions class, often leading to punishments for perceived misbehavior.

This example illustrates how the three themes—scenario, role, and persona—shape the actors’ performances and guide their interactions within a given context. In this study, we have provided detailed definitions of these themes and explored their associations with the concept of social intelligence, demonstrating how they influence the design of AI agents.

4.1.1 Scenario. The dramaturgical theory emphasizes how scenarios, encompassing time, location, and events, work as the “stage” where social interaction occurs [20]. The “stage” establishes the boundaries and expectations for interaction, ensuring that behaviors are meaningful and align with the purpose of the scenarios. This concept is central to the theme of “scenario,” which refers to AI agents designed to operate within a clearly defined application context. Scenarios are intentionally designed to influence or constrain the agents’ actions, providing a framework for their behaviors to be contextually relevant and aligned with specific social norms. The design of scenarios in AI agents can enhance perceived utility and relevance. In contrast, the absence of a scenario indicates that no specific context has been designed to guide or influence the

agent’s behavior, resulting in generalized functionality that lacks adaptation to any particular context.

In our analysis, 28 papers (S1–S28) explicitly integrated scenarios into the agent designs. For example, the agent in S16 was designed to fit a creative writing scenario. Compared with the academic writing scenario, creative writing highlights agents’ behaviors in using specific literary elements and raising attention to clarify ownership and agency issues in generated content [26]. The scenario settings in S16 scope the agent’s actions to sharpen its focus on task-related capabilities, ensuring its functional relevance within the creative writing and reducing requirements for managing hallucinations in large language models. In S22, the researchers created a narrative scenario with conflicts to teach conflict resolution [53]. The designed narrative conflicting scenario in S22 requires the agent to align its behaviors with the scenario’s social norms and the logical progression of the narrative. The agent may begin a conversation in a dissatisfied state, focus on the conflicting topic, and seek a solution. The design of these scenarios includes elements such as time, place, and event, which influence agents’ behavior in various ways.

4.1.2 Role. On the “stage,” actors embody different social identities, each associated with specific expectations and responsibilities that provide a structured framework for behavior in various contexts [20]. The theme “role” refers to the distinct social identity that an AI agent assumes. A role defines the agent’s capabilities and relationship with users, ensuring consistency with social norms and procedural expectations. Conversely, the absence of a defined role results in generalized interactions characterized by unclear responsibilities and ambiguous user expectations.

All 30 included papers described roles assigned to AI agents, highlighting their importance in demonstrating social intelligence. Some papers specified concrete professional roles for the agents, such as an interior designer [27] or a social worker [31]. These roles influenced the agents’ performance by constraining their actions within occupational boundaries to ensure their expertise and behavior align with user expectations, thus enhancing credibility and task relevance. Some papers emphasized fostering specific interactions between agents and users. For example, agents were depicted as collaborators in creating research questions [35] or as a person who sought help from players [66]. These roles shaped the agents’ interaction patterns, focusing on relational dynamics rather than professional attributes, thereby creating meaningful user experiences. In some cases, designers did not explicitly name the role but described agents’ responsibilities, such as focusing on managing and reducing smartphone overuse [44]. The assigned responsibilities influenced the AI’s behavior by establishing clear objectives and guiding actions toward targeted outcomes.

4.1.3 Persona. Even within the same role, actors exhibit distinct characteristics—such as personal traits, behaviors, and communication styles—that shape unique impressions on their audiences [20]. In AI design, this concept extends the expectations associated with a role, ensuring that the agent’s design aligns not only with its functional abilities but also with users’ perceptions of the role, thereby fostering trust and strengthening emotional connections. Based on the widely recognized term in the HCI field that describes the fictitious, specific, concrete representations of target users [38], we label this theme as “persona.” “Persona” examines whether an AI agent is designed with specific characteristics to shape user perceptions, such as personality traits, visual appearance, or communication styles. The absence of a persona results in an AI agent lacking a distinct identity, reducing it to generalized functionality without unique or memorable characteristics.

In our analysis, 15 papers explicitly addressed a persona in their AI agent designs (S1–S13, S29, S30). Some persona designs focus on creating a unique persona to support the design objectives. Designers use predefined or user-customized demographic information, physical appearance, and narrative backgrounds within the agent to create an immersive and engaging experience [45, 54, 61, 64, 66], such as an agent with a realistic background in terms of gender, age, occupation, and education that is shaping its distinct opinion in discussion [64]. Designers also create personas by constraining agent’s communication and interaction strategies. For instance, S4 designed agents to use a simple and clear language style to make its interaction style more suitable for young children [63]. Some personas designs drew on existing personas, such as mythological figures [12], to build intuitive understanding by leveraging familiar traits, cultural recognition, and pre-established behavioral expectations, enhancing user engagement and emotional connection.

4.1.4 Relationship Among Scenario, Role, and Persona. Scenario, role, and persona collectively contribute to exhibiting social intelligence in AI agents by shaping their capabilities, behaviors, and impressions. The scenario provides a specific context that defines the social norms governing interactions within it, setting expectations for how roles should behave. The role specifies the agent’s capabilities and responsibilities, guided by the social norms associated with its identity, which dictate appropriate behaviors. The persona adds a unique impression to the role, infusing it with distinctive characteristics that make the agent’s behavior more relatable and engaging. Together, these themes ensure that an agent’s actions align with user intentions and expectations, thus demonstrating social intelligence.

In addition to defining these themes individually, their collective design should remain consistent to avoid creating agents that deviate from user expectations, even if their behaviors adhere to isolated design elements. For example, the design of a classroom scenario leads to the expectation of presenting roles such as teachers and students, not customer service representatives. Moreover, a teacher’s persona is typically expected to convey knowledgeability and approachability, rather than apathy or incompetence [48]. Misalignment between these themes can affect user trust and hinder establishing positive human-AI relationships.

4.2 Design Implications

Not all included studies incorporated all three design strategies. The choice of which strategies to implement closely aligns with the design objectives, as different combinations cater to varying user expectations across diverse application domains (see table 1).

4.2.1 Design Single-purpose Agents. Some agents are designed to achieve a specific single purpose, focusing on accomplishing design goals that are associated with specific scenarios. The design goal can be primarily task-oriented, with productivity enhancement serving as a key design goal across various fields such as business, service, creativity, and more. Applying scenarios and roles in AI agent design can enable agents to achieve such goals since it provides specific constraints to agents’ abilities and behavior. For example, to support creative interior color design ideation, the designer in S14 assigned the interior design scenario and interior designer role to agents to make it following domain-specific design rationales [27]. The scenario and role design successfully provide a relevant and professional experience for users to improve productivity.

In some cases, besides achieving functional goals, providing an immersive and engaging experience is crucial for designs. For example, in education, engaging learning emphasizes constructing knowledge through interaction [28], requiring agents to engage users and foster meaningful exchanges actively. Our findings suggest integrating scenario, role, and persona into AI agent design created an immersive and engaging experience and demonstrated social intelligence. In S5, designers designed agents as students in the virtual classroom, with the character of being active to break the ice in class, taking notes to encourage others, and being disciplined to remind other students when distraction occurs [36]. Incorporating three elements successfully promotes VR classroom participation in a credible, immersive, and engaging way.

4.2.2 Design Multi-purpose Agents. In some cases, AI agents are designed for various scenarios. The design of such AI agents should ensure the functionality remains consistent and effective while meeting user needs across diverse contexts. The suited role and consistent persona can support achieving this goal. Aligning with the sociological concept of role transitions[2], the role design in this situation should considerate role’s boundary flexibility, which allows a role to be enacted across various environments and times [23]. The inclusion of persona design plays a crucial role in ensuring behavior consistency, which allows the agent to maintain its functional and emotional alignment with users even as scenarios change. For instance, S30 aims to develop agents capable of sustaining health-related disclosures over time [31]. Recognizing the need for conversation across scenarios, the agent was designed with a flexible role, a social worker who engages in casual yet purposeful dialogues. By designing a persona that actively asks health-related questions based on user memory, the agent fosters consistency, trust, and emotional connection, encouraging long-term engagement with socially isolated individuals.

Employing design strategies to enable AI agents to exhibit social intelligence—aligning their actions with user intentions and expectations—is essential for fostering positive human-AI relationships. However, not every design must incorporate all three themes to

Table 1: Overview of design strategies in AI agents across domains. The ✓ indicates the application of certain design strategies.

ID	Scenario	Role	Persona	Domain
S1 [30], S2 [66], S3 [12], S4 [63], S5 [36]	✓	✓	✓	Education
S6 [45], S7 [64], S8 [61], S9 [54]	✓	✓	✓	Simulation
S11 [29], S12 [52], S13 [62]	✓	✓	✓	Other
S10 [22]	✓	✓	✓	Healthcare
S14 [27], S15 [35], S16 [26], S17 [59], S18 [32], S19 [58]	✓	✓	-	Creativity
S20 [10], S21 [11]	✓	✓	-	Education
S22 [53], S23 [25]	✓	✓	-	Simulation
S24 [47], S25 [44], S26 [43]	✓	✓	-	Other
S27 [24], S28 [34]	✓	✓	-	Service
S29 [16]	-	✓	✓	Creativity
S30 [31]	-	✓	✓	Healthcare

achieve social intelligence. The goal of designing socially intelligent AI agents is not to make AI appear human-like but to foster seamless and harmonious interactions between humans and AI. Designers should decide the correct design strategies by clearly understanding their design goals to deliver design solutions that are both efficient and socially intelligent.

5 Limitations and Future Work

This scoping review exclusively focused on papers published in the CHI 2024 proceedings, excluding relevant articles from other databases. While this focus provides a rich collection of HCI-driven AI agent design strategies, it may also limit the diversity of application scenarios identified for the explored themes. Future research could broaden the scope to include additional databases, such as CSCW, IJHCS, and TOCHI, to develop a more comprehensive understanding of the design strategies for socially intelligent AI agents.

Besides, we did not analyze the technical methods used to implement the identified design strategies. As a result, the relationship between design strategies and technical execution remains unexplored. Future work could investigate this relationship, bridging the gap between design and implementation. For instance, it could examine the different effectiveness between technique methods, such as state mechanisms (e.g., Chacha [52]) and prompt engineering (e.g., CloChat [22]), in creating credible personas. Such an exploration would enhance the practical applicability and relevance of these strategies, providing actionable insights for developers and researchers.

Additionally, while our study provides a structured framework for understanding AI agent design strategies, its applicability in real-world AI development has yet to be empirically evaluated. A user study could examine how designers implement this framework in practice and assess its effectiveness in guiding the development of socially intelligent AI agents. Meanwhile, expert interviews could offer insights into its applicability across various domains and identify potential refinements. Future research integrating these empirical methods will be essential for validating and refining the framework.

6 Conclusion

This scoping review examined 30 papers describing the design of AI agents that demonstrate social intelligence. Through our analysis,

we identified recurring patterns in design strategies, categorized into three themes: scenario, role, and persona. We explored how these themes relate to specific design objectives and their implications for creating socially intelligent AI agents. Our findings highlight that demonstrating social intelligence in AI is less about adhering to a full-spectrum design and more about enabling the agent to meet human expectations in contextually appropriate ways. By carefully tailoring design strategies to align with specific goals and social norms, AI agents can bridge the gap between human expectations and AI behaviors, fostering harmonious and positive relationships.

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Table 2: Design Strategies in AI Agents of the Recent Research

ID	Scenario	Role	Persona	Domain
S1 [30]	Programming education	Tutees: a 2nd-year high school student	Ask "Why & How"; specific levels of knowledge and misconceptions	Education
S2 [66]	Distant land that has endured severe climate-related devastation	Memory-impaired individual that seeking for users help	Memory-impaired, alien, etc.	Education
S3 [12]	AR educational activities with a narrative inspired by Chinese mythology	Corresponds with questions and answers	Character from Chinese mythology: Chang'e	Education
S4 [63]	Develop math language through story-telling for children aged 4-8	Story co-creator	Use simple, clear language that young children can easily understand	Education
S5 [36]	Virtual reality classroom	Virtual students for promoting classroom participation	Active - break the ice, taking notes ; disciplined - distraction reminder	Education
S6 [45]	Customized narrative story	Customized role	Customized character	Simulation
S7 [64]	Comments section of the online forum	Discussing opinions	Realistic background in terms of gender, age, etc.	Simulation
S8 [61]	Definitive usage scenarios	Simulated user	User ability characteristics and task-related characteristics	Simulation
S9 [54]	Online public video platforms	Debate partners	Social identity and rhetorical styles	Simulation
S10 [22]	12 Pre-designed scenarios	Customized role	Customized demographic information, verbal and nonverbal style, appearance, etc.	Other
S11 [29]	Workplace	Workplace-communication assistant	Workplace appropriate style; short answer	Healthcare
S12 [52]	Prompt children to share emotions about personal events	A facilitator between children and parents	Peer child	Healthcare
S13 [62]	Support people living with dementia for reminiscence	Facilitate discussions around the memories linked to photos	Gender, voice accent, and memory of users uploaded photos	Healthcare
S14 [27]	Interior design	Interior designer	-	Creativity
S15 [35]	Creating research questions	Collaborator	-	Creativity
S16 [26]	Creative writing	Assistant that managing interaction among writers and LLMs	-	Creativity
S17 [59]	Translate news articles into video scripts and storyboards	Similar to "apprentice/assistant" within the apprentice framework	-	Creativity
S18 [32]	Diary writing	Writing assistant	-	Creativity
S19 [58]	Dream journal	Writing assistant (generates visual metaphors and text depictions)	-	Creativity
S20 [10]	Learn NetLogo, an agent-based modeling language	Assistant and collaborator	-	Education
S21 [11]	Programming learning for children aged 6-12	Educator who guide children with stories and assistant	-	Education
S22 [53]	Specific conflicting situation	Simulated user and expert	-	Simulation
S23 [25]	Specific cyberbullying situation	Teachers and simulated students	-	Simulation
S24[47]	Workplace environment with tasks that requires two to complete collaboratively	Collaborator	-	Other
S25 [44]	Overuse smartphone	Manage and reduce smartphone overuse	-	Other
S26 [43]	Assist blind or low vision people to explore video content	Describe video content	-	Other
S27 [24]	VR shopping	Shopping assistant that recommend products	-	Service
S28 [34]	Waiting room in hospitals	Patient-intake bot: physician assistant	-	Service
S29 [16]	-	Topic agent and artist agent	Knows key preferences of the user	Creativity
S30 [31]	-	Social worker who check in with socially isolated individuals	Memory health related content; ask for updates on previous issues; casual conversations	Healthcare